



DIRECT MEMORY ACCESS EVERYWHERE

securelyfitz & baelfire 9 AUG 2025

JOE FITZPATRICK

- 10+ years of fun with hardware
 - silicon debug
 - security research
 - pen testing of CPUs
 - security training
- Applied Physical Attacks Training:
 - X86 Systems
 - Embedded Systems
 - Hardware Pentesting
 - Own white shoes full of LEDs



Joe FitzPatrick

@securelyfitz

joefitz@securinghardware.com

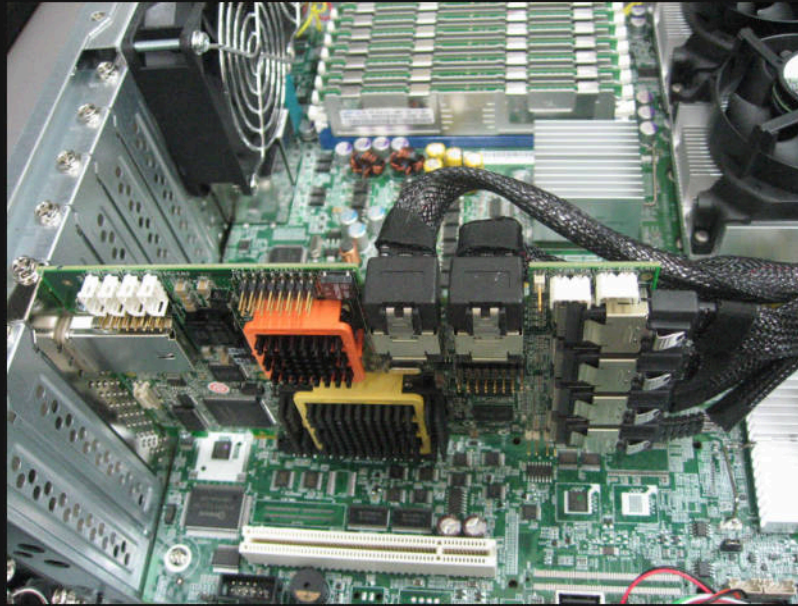
GRACE PARRISH

- Studying cybersecurity at OSU
- Working as SOC analyst
- Board level repair technician
- Taught industrial control systems 9 years
 - PLCs
 - HMIs
- Enjoys learning
 - Jack of all trades
- Goal: offensive security career

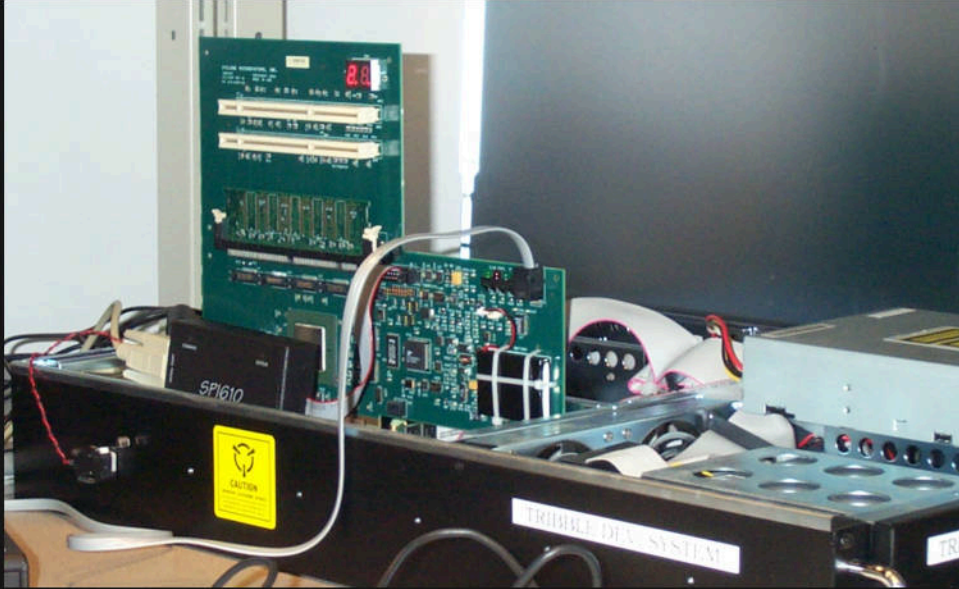


Grace Parrish

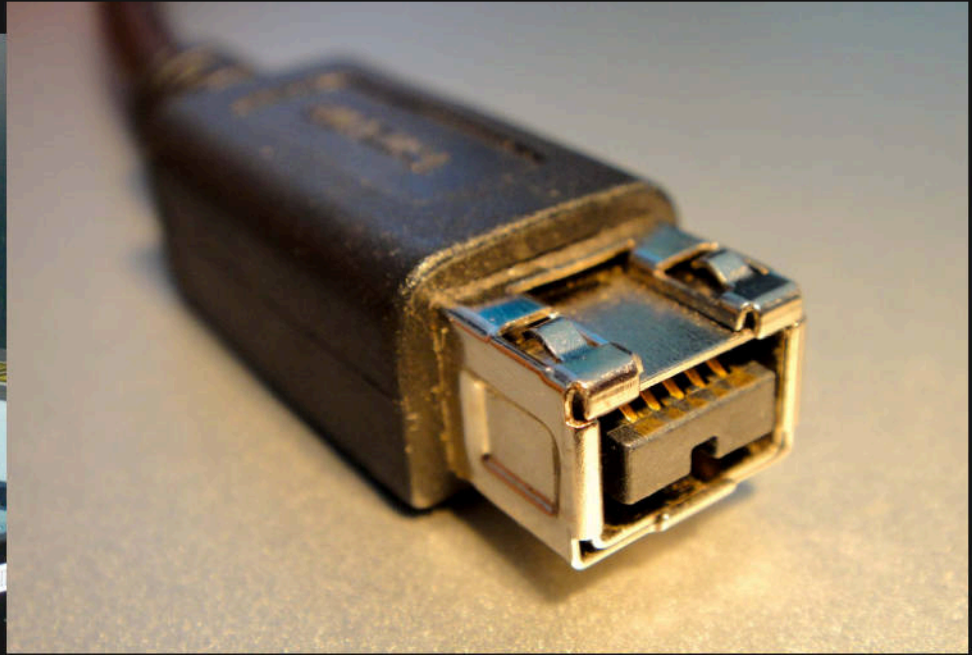
@BaelfireNightshd



This is PCIe.



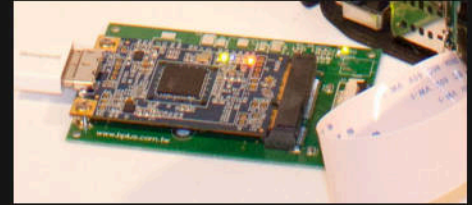
Tribble, a PCI DMA attack device by Joe Grand



Firewire, external dma spanning 2 millenia

A brief tale of epic procrastination:

- 2014: Slotscreamer
- 2016: Pcileech
- 2017: Pcileech FPGA
- 2019: Thunderclap
- 2020: Thunderspy
- more to come...



4 QUESTION THREAT MODEL:

- 1.
- 2.
- 3.
- 4.

4 QUESTION THREAT MODEL:

1. What are we working on?
- 2.
- 3.
- 4.

4 QUESTION THREAT MODEL:

1. What are we working on?
2. What can go wrong?
- 3.
- 4.

4 QUESTION THREAT MODEL:

1. What are we working on?
2. What can go wrong?
3. What are we going to do about it?
- 4.

4 QUESTION THREAT MODEL:

1. What are we working on?
2. What can go wrong?
3. What are we going to do about it?
4. Did we do a good job?

PCIE THREAT MODEL

1. What are we working on?
2. What can go wrong?
3. What are we going to do about it?
4. Did we do a good job?

PCIE THREAT MODEL

1. What are we working on?

High Speed interconnects

2. What can go wrong?

3. What are we going to do about it?

4. Did we do a good job?

PCIE THREAT MODEL

1. What are we working on?

High Speed interconnects

2. What can go wrong?

Malicious Devices

3. What are we going to do about it?

4. Did we do a good job?

PCIE THREAT MODEL

1. What are we working on?

High Speed interconnects

2. What can go wrong?

Malicious Devices

3. What are we going to do about it?

Keep it inside the case

4. Did we do a good job?

PCIE THREAT MODEL

1. What are we working on?

High Speed interconnects

2. What can go wrong?

Malicious Devices

3. What are we going to do about it?

Keep it inside the case

4. Did we do a good job?

Mostly?

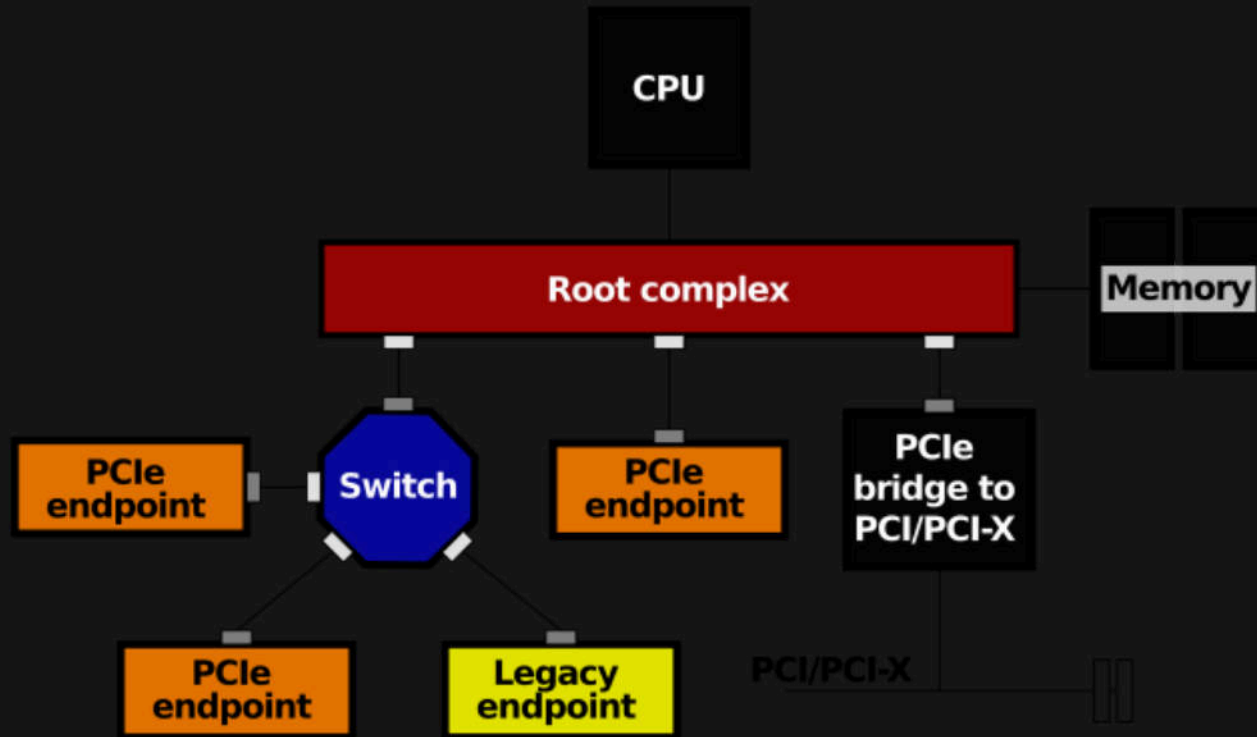
ATTACKING PCIE

1. Connect your hardware
2. Enumerate
3. Enable Bus Master
4. Read or Write memory
5. Present at DEF CON

CONNECTING HARDWARE



ENUMERATING



ENUMERATING

```
joefitz@linmax2:~$ lspci -tv
```

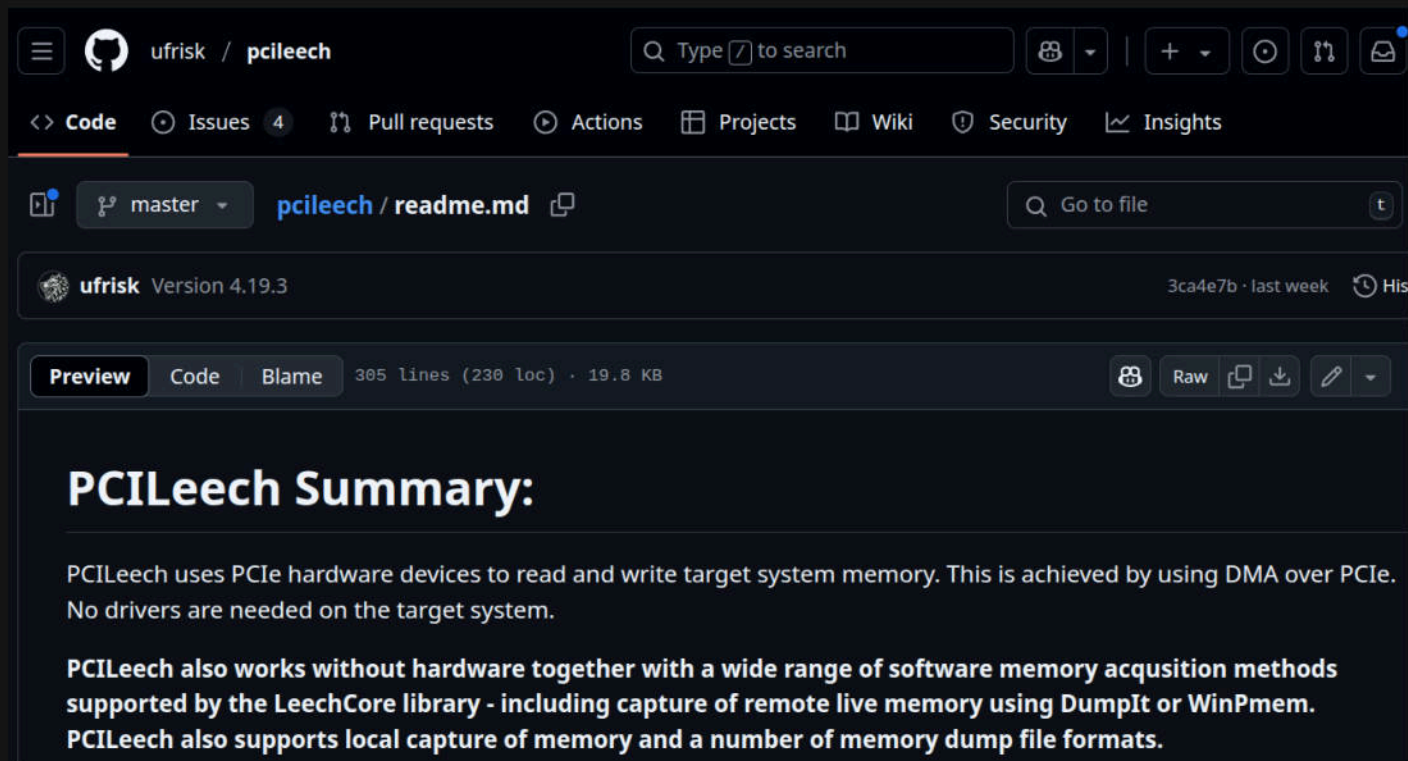
```
-[0000:00]-+-00.0  Advanced Micro Devices, Inc. [AMD] Strix/Strix Halo Root Complex
+-00.2  Advanced Micro Devices, Inc. [AMD] Strix/Strix Halo IOMMU
+-01.0  Advanced Micro Devices, Inc. [AMD] Strix/Strix Halo Dummy Host Bridge
+-01.1-[01-60]--
+-01.2-[61-c0]--
+-02.0  Advanced Micro Devices, Inc. [AMD] Strix/Strix Halo Dummy Host Bridge
+-02.1-[c1]----00.0  Shenzhen Longsys Electronics Co., Ltd. Lexar NM790 NVME SSD (DRAM-less)
+-02.2-[c2]----00.0  Genesys Logic, Inc GL9755 SD Host Controller
+-02.3-[c3]----00.0  Intel Corporation Wi-Fi 6E(802.11ax) AX210/AX1675* 2x2 [Typhoon Peak]
+-03.0  Advanced Micro Devices, Inc. [AMD] Strix/Strix Halo Dummy Host Bridge
+-08.0  Advanced Micro Devices, Inc. [AMD] Strix/Strix Halo Dummy Host Bridge
```

READ/WRITE MEMORY

Table 2-2: PCI Express TLP Packet Types

TLP Packet Types	Abbreviated Name
Memory Read Request	MRd
Memory Read Request - Locked access	MRdLk
Memory Write Request	MWr
IO Read	IORd
IO Write	IOWr
Configuration Read (Type 0 and Type 1)	CfgRd0, CfgRd1
Configuration Write (Type 0 and Type 1)	CfgWr0, CfgWr1
Message Request without Data	Msg
Message Request with Data	MsgD
Completion without Data	Cpl
Completion with Data	CplD
Completion without Data - associated with Locked Memory Read Requests	CplLk
Completion with Data - associated with Locked Memory Read Requests	CplDLk

PCILEECH



ufrisk / pcileech

Q Type / to search

<> Code Issues 4 Pull requests Actions Projects Wiki Security Insights

master pcileech / readme.md

Go to file

ufrisk Version 4.19.3 3ca4e7b · last week His

Preview Code Blame 305 lines (230 loc) · 19.8 KB

Raw

PCILeech Summary:

PCILeech uses PCIe hardware devices to read and write target system memory. This is achieved by using DMA over PCIe. No drivers are needed on the target system.

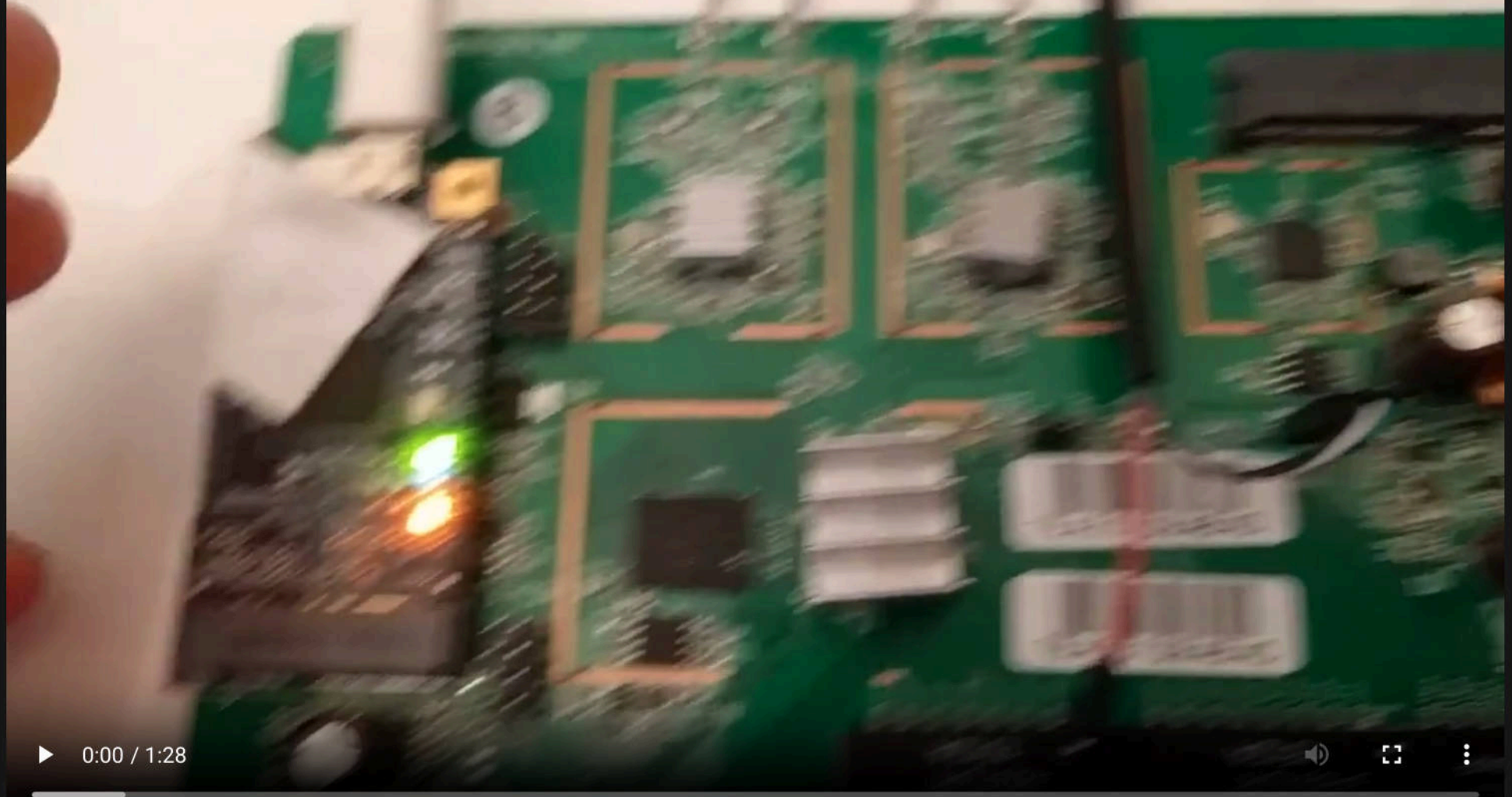
PCILeech also works without hardware together with a wide range of software memory acquisition methods supported by the LeechCore library - including capture of remote live memory using DumpIt or WinPmem. PCILeech also supports local capture of memory and a number of memory dump file formats.

EXAMPLE: DUMPING MEMORY



EXAMPLE: LOCKSCREEN BYPASS





▶ 0:00 / 1:28



KEY TAKEAWAY:

PCIE IS PERMISSIVE BY DESIGN

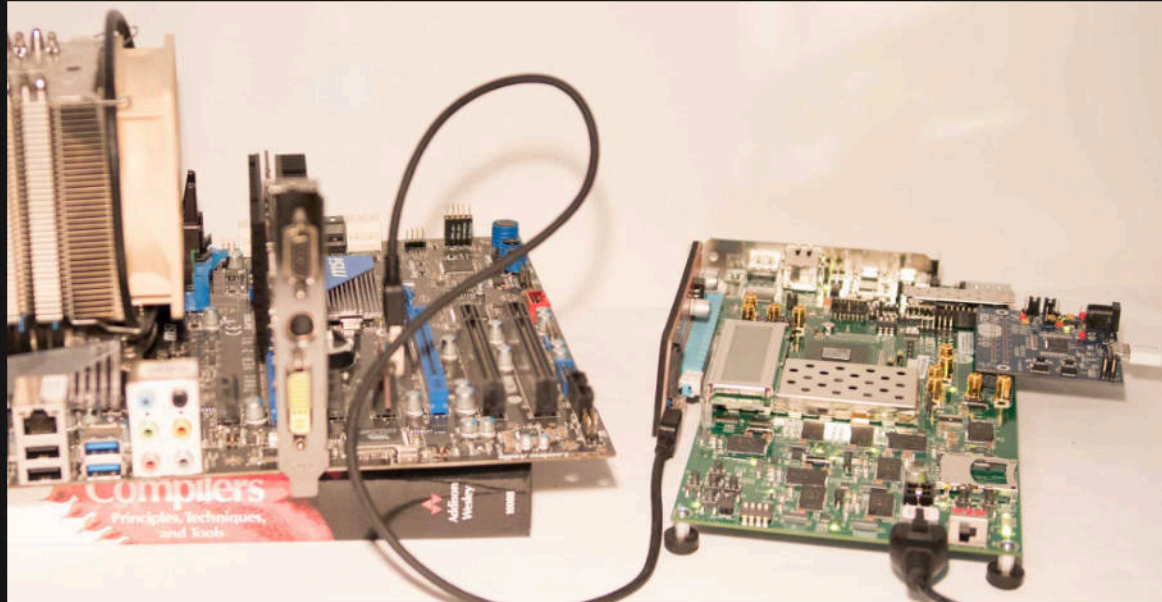
Joe's DMA Attack Taxonomy:

- Tethered
- Drive-By
- Embedded

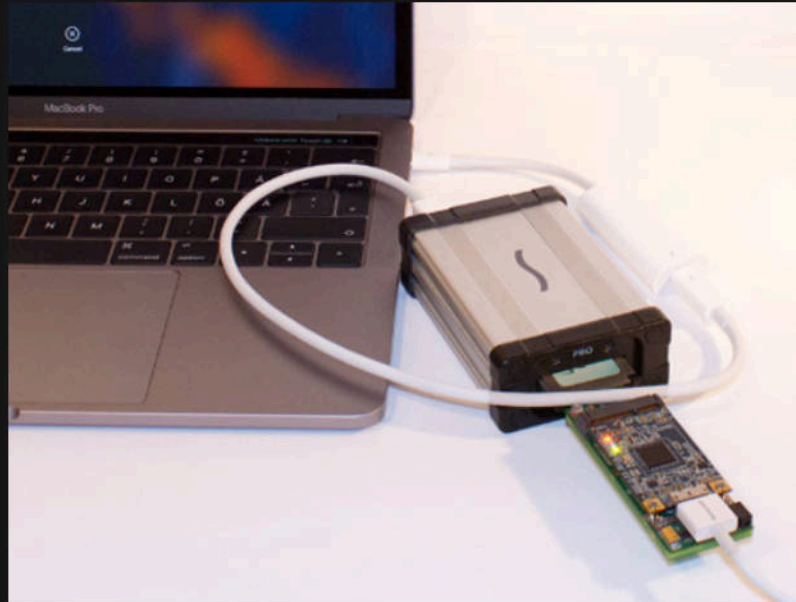


FYI: I made this up

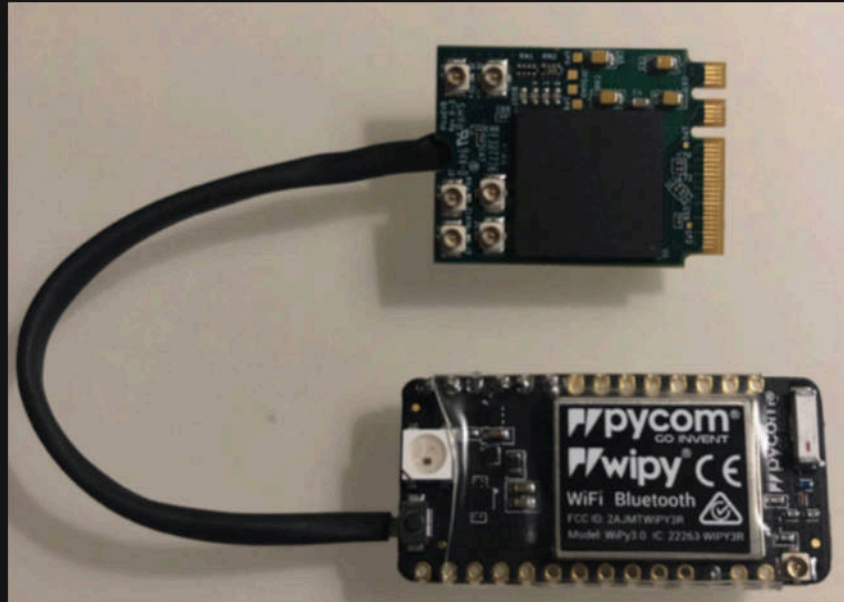
TETHERED



DRIVE-BY



EMBEDDED



Types of DMA Attacks

	Short Term	Long Term
Quick access/ hot plug	Drive By	Tethered
Internal access/ reboot	Tethered	Embedded

MULTITENANT THREAT MODEL

1. What are we working on?
2. What can go wrong?
3. What are we going to do about it?
4. Did we do a good job?

MULTITENANT THREAT MODEL

1. What are we working on?

Make \$ by sharing hardware

2. What can go wrong?

3. What are we going to do about it?

4. Did we do a good job?

MULTITENANT THREAT MODEL

1. What are we working on?

Make \$ by sharing hardware

2. What can go wrong?

Hardware assisted tenant hopping

3. What are we going to do about it?

4. Did we do a good job?

MULTITENANT THREAT MODEL

1. What are we working on?

Make \$ by sharing hardware

2. What can go wrong?

Hardware assisted tenant hopping

3. What are we going to do about it?

IOMMU!

4. Did we do a good job?

MULTITENANT THREAT MODEL

1. What are we working on?

Make \$ by sharing hardware

2. What can go wrong?

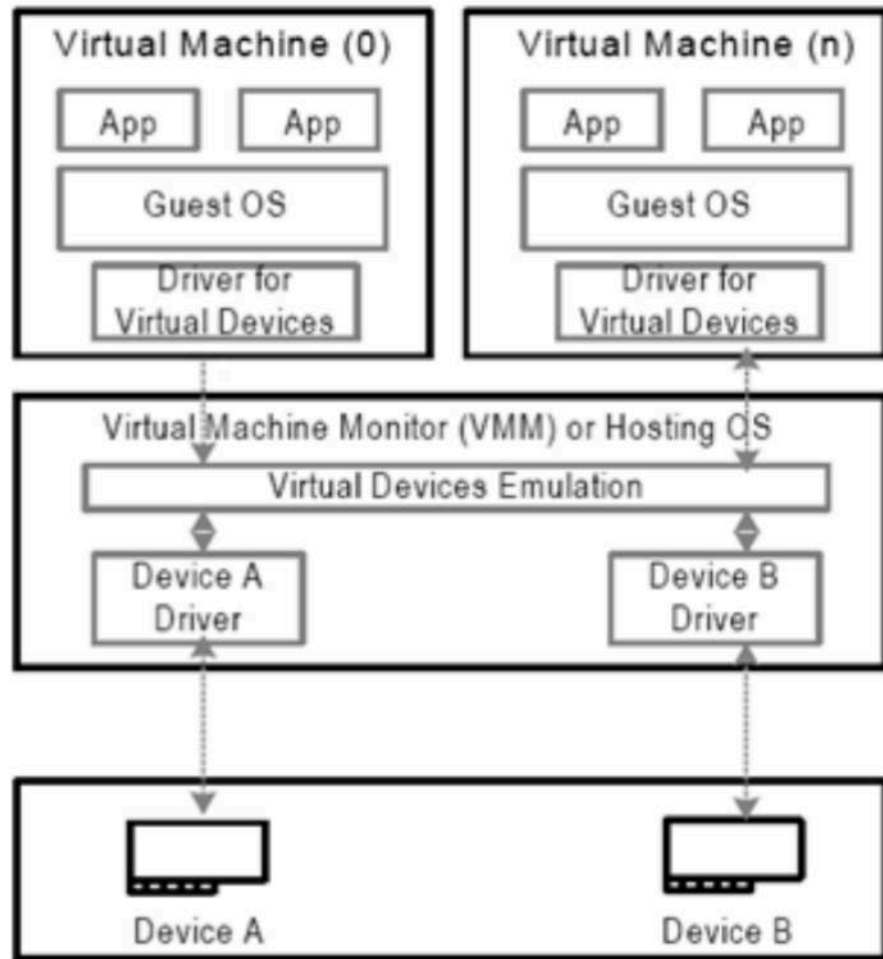
Hardware assisted tenant hopping

3. What are we going to do about it?

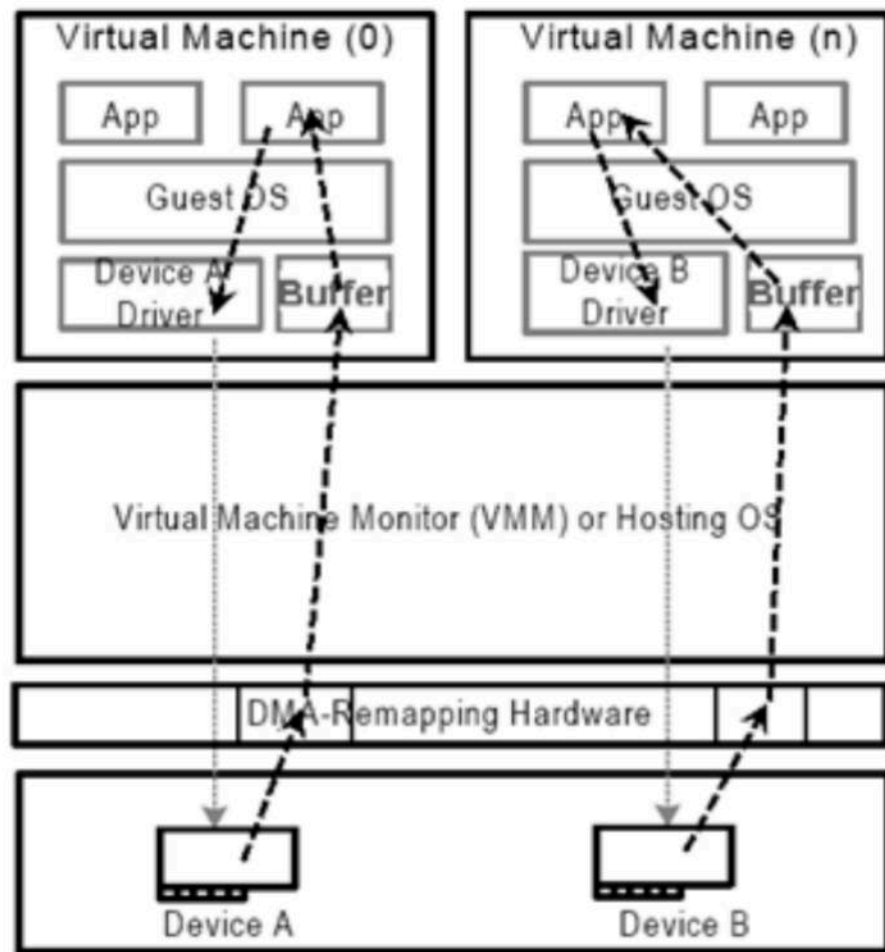
IOMMU!

4. Did we do a good job?

Mostly?



Example Software-based
I/O Virtualization



Direct Assignment of I/O Devices

KEY TAKEAWAY:
DMA MITIGATIONS EXIST

THUNDERBOLT THREAT MODEL

1. What are we working on?
2. What can go wrong?
3. What are we going to do about it?
4. Did we do a good job?

THUNDERBOLT THREAT MODEL

1. What are we working on?

High speed peripherals

2. What can go wrong?

3. What are we going to do about it?

4. Did we do a good job?

THUNDERBOLT THREAT MODEL

1. What are we working on?

High speed peripherals

2. What can go wrong?

Malicious devices

3. What are we going to do about it?

4. Did we do a good job?

THUNDERBOLT THREAT MODEL

1. What are we working on?

High speed peripherals

2. What can go wrong?

Malicious devices

3. What are we going to do about it?

Authenticate and Isolate devices

4. Did we do a good job?

THUNDERBOLT THREAT MODEL

1. What are we working on?

High speed peripherals

2. What can go wrong?

Malicious devices

3. What are we going to do about it?

Authenticate and Isolate devices

4. Did we do a good job?

In time... yes!

About those mitigations:

- Thunderbolt released
- OSX 10.8.2
- Windows 10 1802
- Linux 5.0

About those mitigations:

- Thunderbolt released (*2011*)
- OSX 10.8.2
- Windows 10 1802
- Linux 5.0

About those mitigations:

- Thunderbolt released (*2011*)
- OSX 10.8.2 (*2012, +1 year*)
- Windows 10 1802
- Linux 5.0

About those mitigations:

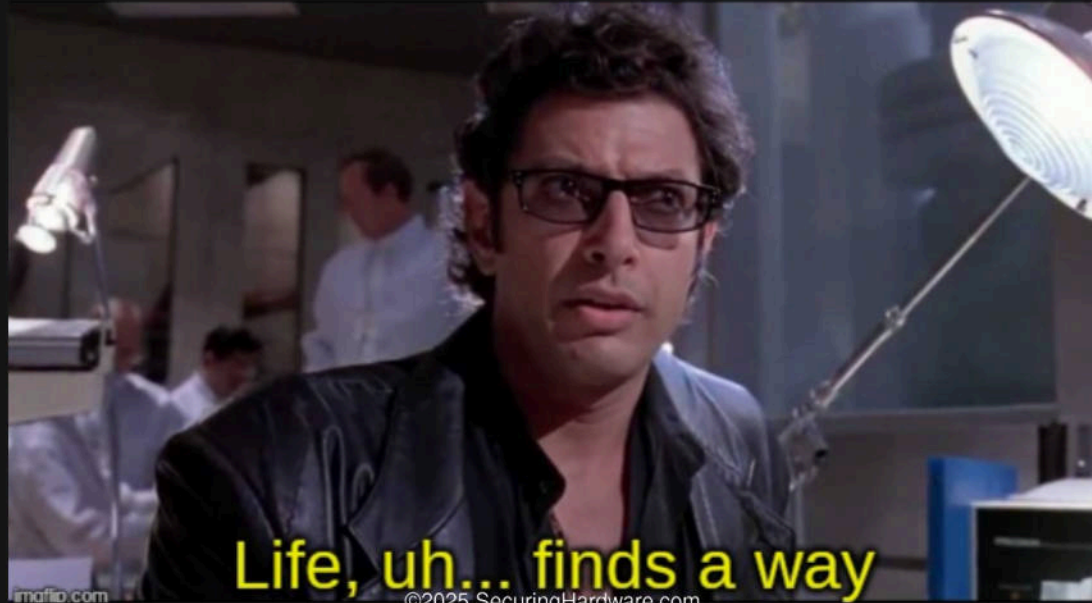
- Thunderbolt released (*2011*)
- OSX 10.8.2 (*2012, +1 year*)
- Windows 10 1802 (*2018, + 7 years*)
- Linux 5.0

About those mitigations:

- Thunderbolt released (*2011*)
- OSX 10.8.2 (*2012, +1 year*)
- Windows 10 1802 (*2018, + 7 years*)
- Linux 5.0 (*2019, + 8 years*)

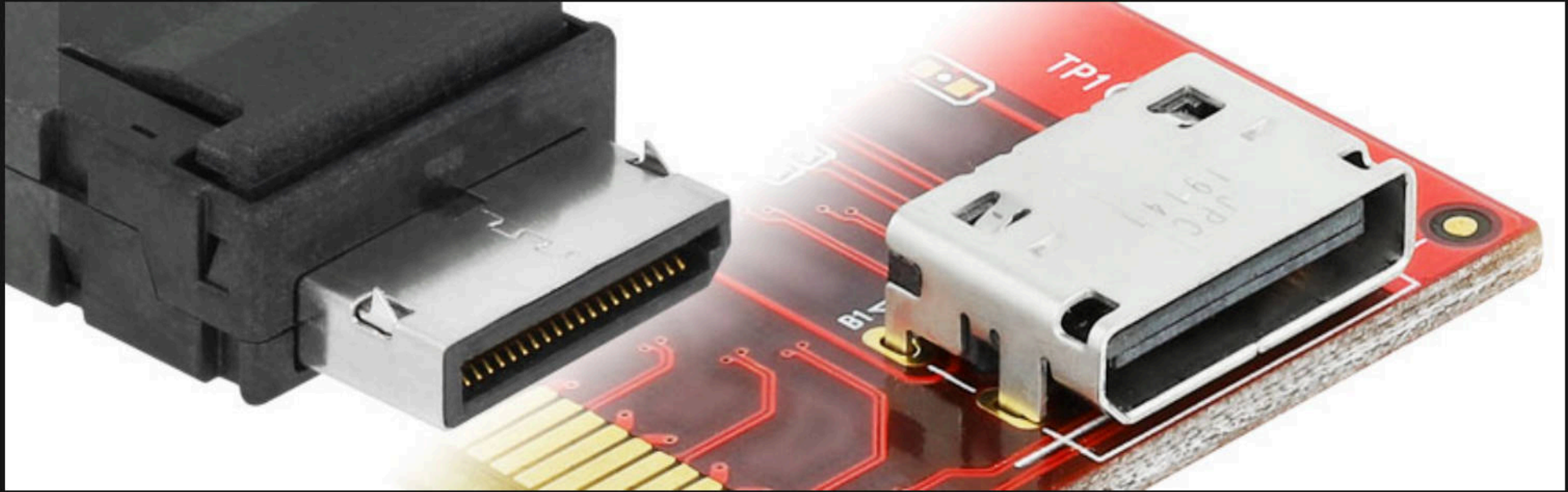
KEY TAKEAWAY:
THUNDERBOLT GOT FIXED, EVENTUALLY

HOW ARE WE GONNA DO DRIVE-BY DMA ATTACKS WHEN THUNDERBOLT IS AT LEAST SOMEWHAT SECURED?



imafio.com

©2025 SecuringHardware.com





Tigard

OCULINK THREAT MODEL

1. What are we working on?
2. What can go wrong?
3. What are we going to do about it?
4. Did we do a good job?

OCULINK THREAT MODEL

1. What are we working on?

High speed peripherals

2. What can go wrong?

3. What are we going to do about it?

4. Did we do a good job?

OCULINK THREAT MODEL

1. What are we working on?

High speed peripherals

2. What can go wrong?

Malicious devices

3. What are we going to do about it?

4. Did we do a good job?

OCULINK THREAT MODEL

1. What are we working on?

High speed peripherals

2. What can go wrong?

Malicious devices

3. What are we going to do about it?

Only use it in secured locations?

4. Did we do a good job?

OCULINK THREAT MODEL

1. What are we working on?

High speed peripherals

2. What can go wrong?

Malicious devices

3. What are we going to do about it?

Only use it in secured locations?

4. Did we do a good job?

Not really.

Example: Metamask

The screenshot shows the Firefox Add-ons page for the 'MetaMask – Crypto Wallet' extension. The browser's address bar shows the URL `addons.mozilla.org/en-US/firefox/addon/ether-metamask/?utm_source=www.google.com`. The page header includes the Firefox logo and the text 'Firefox Browser ADD-ONS'. Below the header, there is a warning message: 'This add-on is not actively monitored for security by Mozilla. Make sure you trust it before installing'. The extension's name 'MetaMask – Crypto Wallet' is displayed, along with the developer 'by danfinlay'. A star rating of 4.1 (4,479 reviews) and a user count of 382,986 Users are shown. An 'Add MetaMask' dialog box is open on the right, listing required permissions: 'Access your data for all websites', 'Input data to the clipboard', and 'Display notifications to you'. It also includes an optional setting: 'Allow extension to run in private windows'. The dialog has 'Cancel' and 'Add' buttons.

MetaMask – Crypto Wallet
by danfinlay

The most secure wallet for crypto, NFTs, and DeFi, trusted by millions of users

★ 4.1 (4,479 reviews) 382,986 Users

Add MetaMask

Required permissions:

- Access your data for all websites
- Input data to the clipboard
- Display notifications to you

Optional settings:

☐ Allow extension to run in private windows

[Learn more](#)

Cancel Add

▶ 0:00 / 1:20

©2025 SecuringHardware.com



<https://github.com/pierce403/PASIV>

Preview Code Blame 102 lines (74 loc) · 4.87 KB

PASIV: Peripheral Attack Simulation and Introspection Vehicle

PASIV is a comprehensive test harness for emulating, executing, and analyzing Direct Memory Access (DMA) attacks within a controlled QEMU/KVM environment. This framework aims to provide researchers and security professionals with a powerful, scriptable, and repeatable platform for studying DMA vulnerabilities and defenses.

Project Goals

- Automate the setup of vulnerable and hardened virtual machine targets.
- Integrate with leading memory forensics tools like PCILeech, memflow, and MemProcFS.
- Provide a modular framework for developing and executing custom DMA attack payloads.
- Enable the study and verification of IOMMU-based defenses.
- Serve as a research platform for advanced DMA attack and defense techniques.



DMA GAMING CHEATS

Now, the attacker is the owner.

This changes the threat model!

This is currently an arms race!

DMA cheat for Faceit CS2

Only cheat



\$30

- ✓ CS2 cheat for 1 month ⓘ

* you should have DMA Card with Firmware to use this cheat

Buy now

Basic package



\$720

- ✓ CS2 cheat for 1 month ⓘ
- ✓ DMA PCIe Card 75T ⓘ
- ✓ Firmware for Faceit

* with this package you will be able to use only radar hack on second PC

Out of stock

Full package



\$930

- ✓ CS2 cheat for 1 month ⓘ
- ✓ DMA PCIe Card 75T ⓘ
- ✓ Firmware for Faceit
- ✓ Makcu for Aimbot ⓘ
- ✓ DP Fuser for ESP on monitor ⓘ

Out of stock

Ptolemaios9 Update README.md

97e9651 · 5 days ago History

Preview

Code

Blame

21 lines (15 loc) · 1.57 KB



Raw



Pcileech-DMA-NVMe-VMD

👤 Many firmware scammers sell Beaters free and open source VMD firmware at high prices.

How to get help

discord : <https://discord.gg/beater>

🔥 Beaters DC channel offers free firmware (Many scammers firmware distributors are angry because Beater has made VMD firmware available for free, Try to discredit Beater by claiming that the free firmware is unsafe so that users can buy their paid Poor quality firmware.),BeaterFreeVMD play the games you want to play!VGK has implemented VMD detection not long ago, and EAC BE still runs well. Beater Never tried to sell VMD firmware solutions for profit.

Requirements↓

- Intel CPU (11th Generation or newer) in the main PC where the DMA card is installed.
- Intel VMD (Virtual RAID on CPU) must be enabled in BIOS.
- Specific Intel drivers must be installed on Windows.
- A Windows reinstall may be required for proper driver initialization and device recognition.

⚠️VMD firmware solution has been discovered by game developers, but it is still safe, It has been free and open source from the beginning.

Hardware based memory acquisition methods:

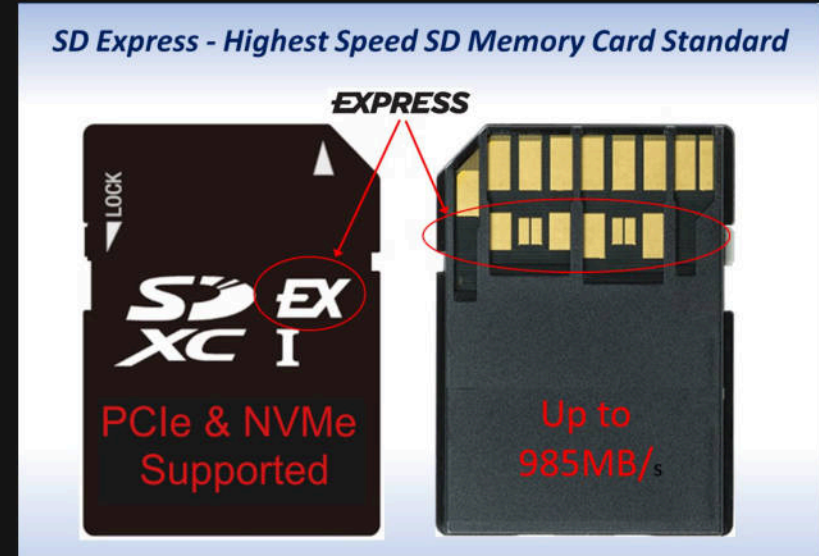
Please find a summary of the supported hardware based memory acquisition methods listed below. All hardware based memory acquisition methods are supported on both Windows and Linux. The FPGA based methods however sports a slight performance penalty on Linux and will max out at approx: 90MB/s compared to 150MB/s on Windows.

Device	Type	Interface	Speed	64-bit memory access	PCIe TLP access	Project Sponsor
Screamer PCIe Squirrel	FPGA	USB-C	190MB/s	Yes	Yes	💖
ZDMA	FPGA	Thunderbolt3	1000MB/s	Yes	Yes	💖
LeetDMA	FPGA	USB-C	190MB/s	Yes	Yes	💖
AC701/FT601	FPGA	USB3	190MB/s	Yes	Yes	
USB3380-EVB	USB3380	USB3	150MB/s	No	No	
DMA patched HP iLO	BMC	TCP	1MB/s	Yes	No	

KEY TAKEAWAY:
CHANGING FORM FACTOR
=
CHANGING THREAT MODEL



CF Express



SD Express

CFE/SDE THREAT MODEL

1. What are we working on?
2. What can go wrong?
3. What are we going to do about it?
4. Did we do a good job?

CFE/SDE THREAT MODEL

1. What are we working on?

High speed small storage

2. What can go wrong?

3. What are we going to do about it?

4. Did we do a good job?

CFE/SDE THREAT MODEL

1. What are we working on?

High speed small storage

2. What can go wrong?

Malicious devices? Counterfeit devices?

3. What are we going to do about it?

4. Did we do a good job?

CFE/SDE THREAT MODEL

1. What are we working on?

High speed small storage

2. What can go wrong?

Malicious devices? Counterfeit devices?

3. What are we going to do about it?

???

4. Did we do a good job?

CFE/SDE THREAT MODEL

1. What are we working on?

High speed small storage

2. What can go wrong?

Malicious devices? Counterfeit devices?

3. What are we going to do about it?

???

4. Did we do a good job?

???



EXAMPLE: CFEXPRESS ATTACK



EXAMPLE: CFEXPRESS ATTACK



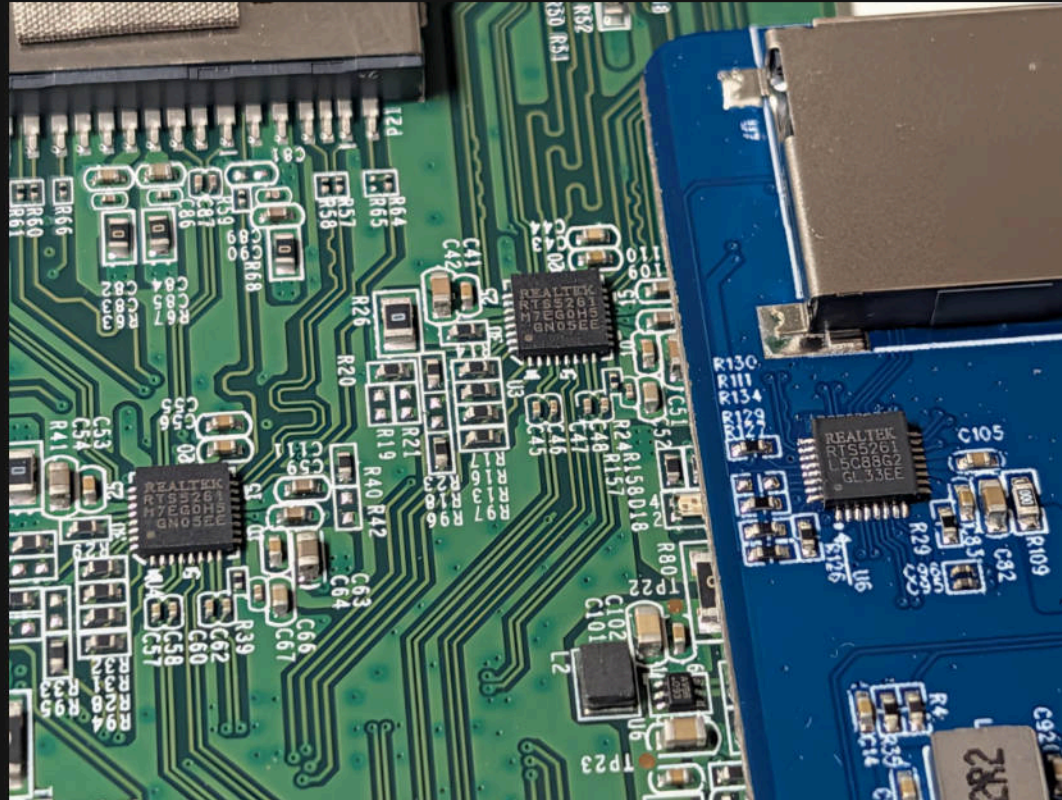
EXAMPLE: CFEXPRESS ATTACK



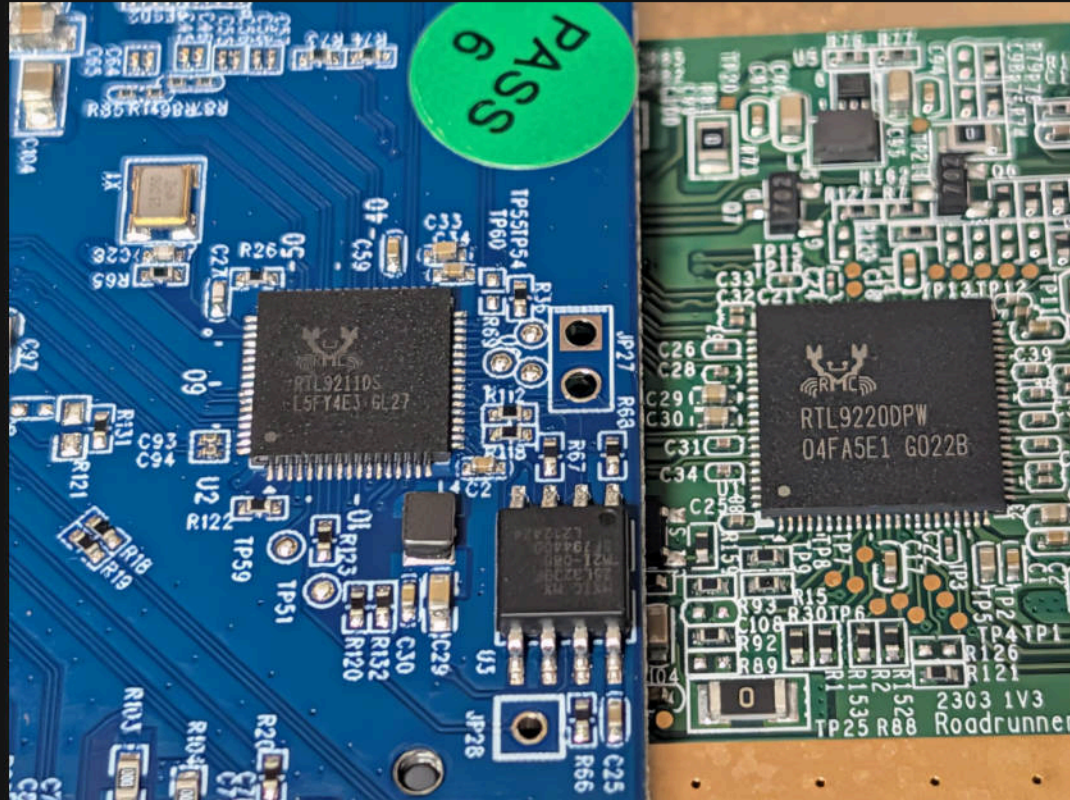
SWITCH 2 SDEXPRESS PROGRESS



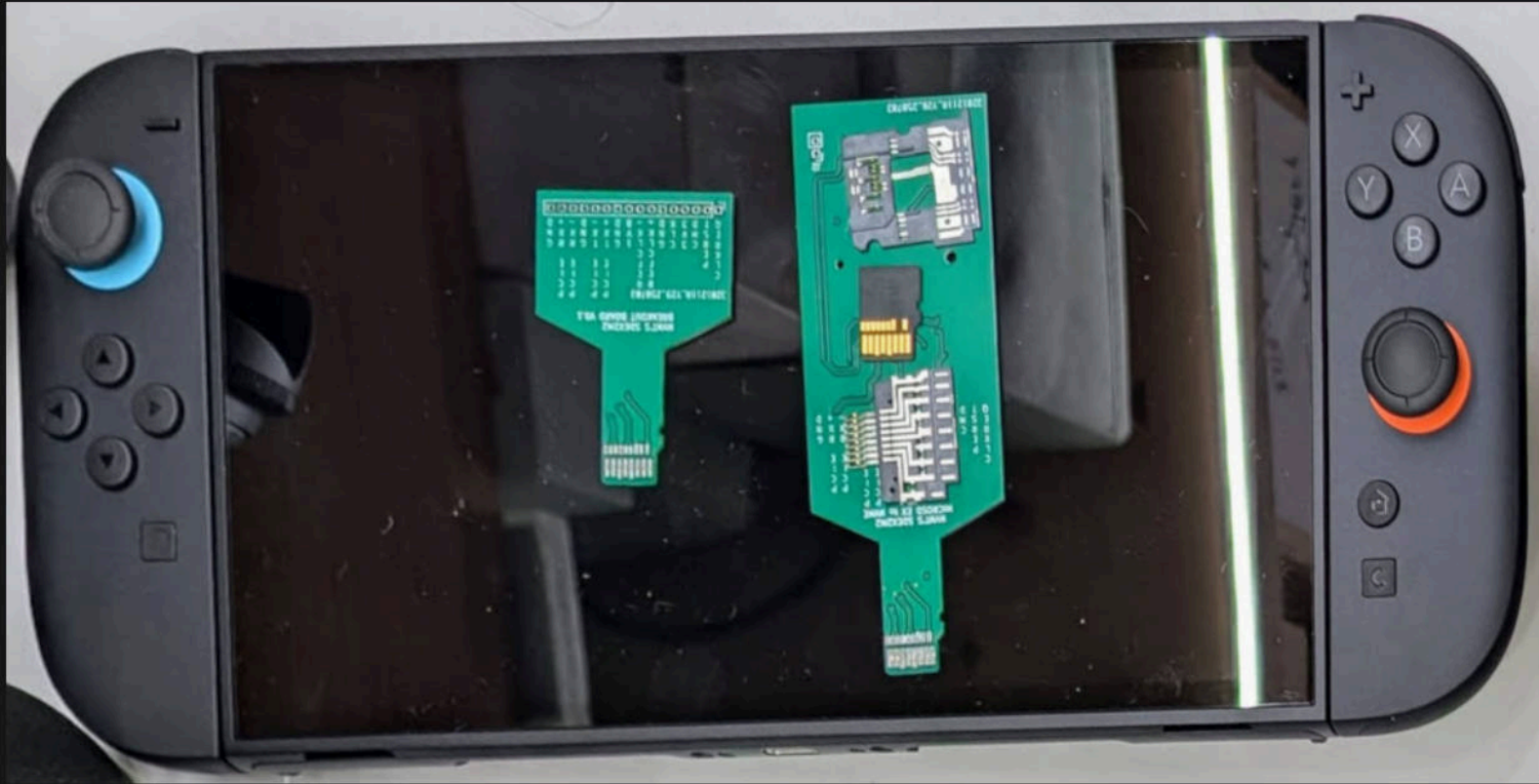
SDExpress Progress



SDExpress Progress



Who wants to see a switch 2 jailbreak?

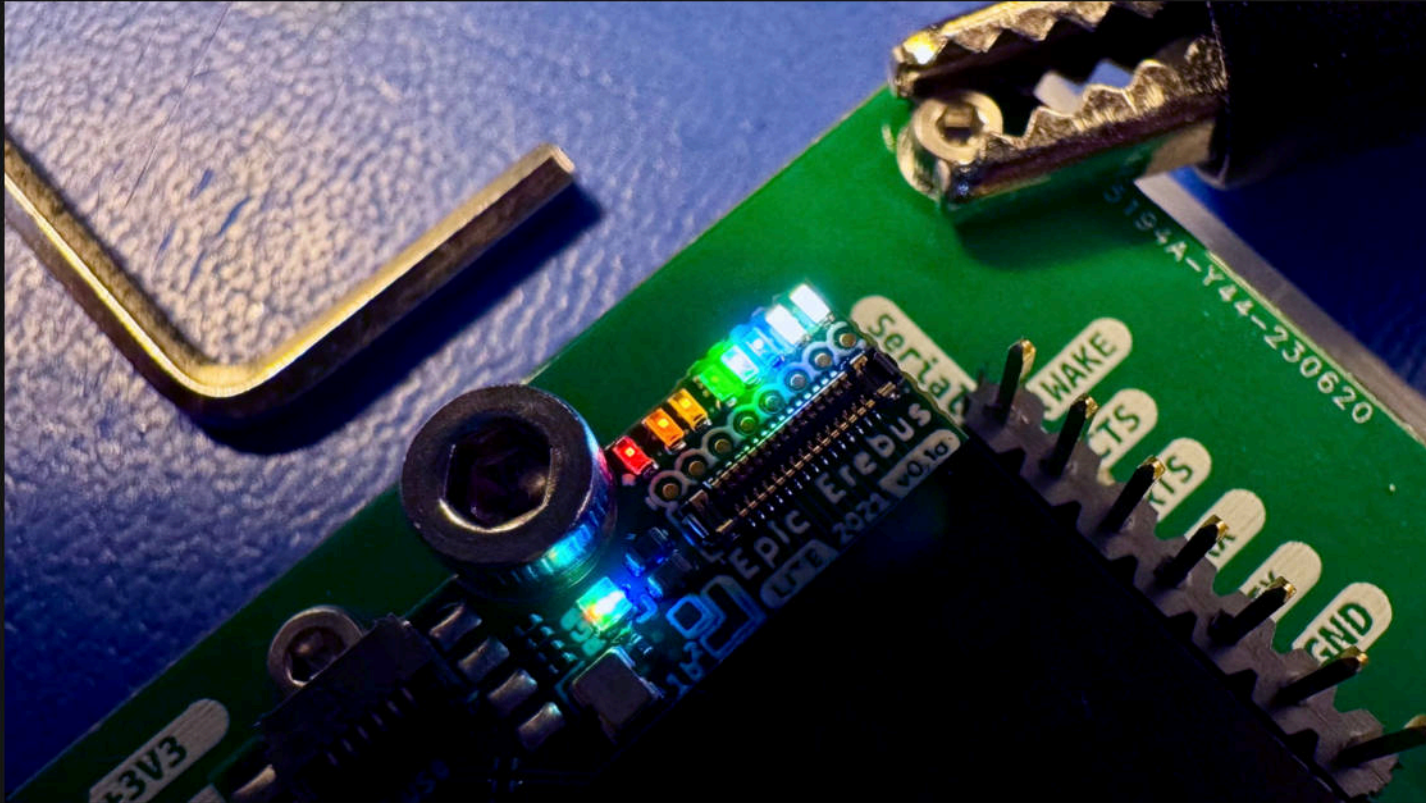


KEY TAKEAWAY:
PCIE IS PERVASIVE!

A CONTINUING TALE OF EPIC PROCRASTINATION:

- 2021: Erebus Schematic
- 2022: Erebus PCBs
- 2023: Erebus Test Jig
- 2024: Erebus Assembled
- 2025: Profit?





Other Notes:

The completed solution contains Xilinx proprietary IP cores licensed under the Xilinx CORE LICENSE AGREEMENT. This project as-is published on Github contains no Xilinx proprietary IP. Published source code are licensed under the MIT License. The end user that have downloaded the no-charge Vivado WebPACK from Xilinx will have the proper licenses and will be able to re-generate Xilinx proprietary IP cores by running the build detailed above.

Projects Launched



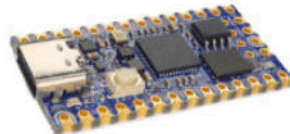
EPIC Erebus

A tiny PCIe DMA tool that's fully customizable with an open toolchain and gateway

Coming Soon

Sign up

Subscribe for project updates.



iCEBreaker-bitsy FPGA

An open source iCE40 FPGA dev board in a Teensy form factor

Coming Soon

Sign up

Subscribe for project updates.



iCEBreaker FPGA

An open source iCE40 FPGA development board designed for teachers and students

992% Funded!

In Stock

\$148,949
raised

14
updates

1,131
backers



Tigard

An open source FT2232H-based, multi-protocol, multi-voltage tool for hardware hacking

Funded!

In Stock

\$145,610
raised

14
updates

1,480
backers



Glasgow Interface Explorer

A highly capable and extremely flexible open source multitool for digital electronics

1,720% Funded!

In Stock

\$430,222
raised

14
updates

2,212
backers

© 2025 Seeed Studio Hardware

WHY EREBUS?

Platform Security

KEY TAKEAWAYS:

- PCIe is permissive by design
- DMA Mitigations exist
- Thunderbolt got fixed, eventually
- Form factor changes your threat model
- PCIe is Pervasive!

Progress has been made -

but there's still way more to do!

4 QUESTION THREAT MODEL:

1. What are we working on?
2. What can go wrong?
3. What are we going to do about it?
4. Did we do a good job?



[REGISTER NOW](#)

Sep 30 - Oct 2, 2025

METRO TORONTO CONVENTION CENTRE

[ATTEND](#) ▾

[BRIEFINGS](#) ▾

[ARSENAL](#) ▾

[SUMMITS](#) ▾

[FEATURES](#) ▾

[BUSINESS HALL](#)

[SPONSORS](#) ▾

All times are in Eastern Time (GMT/UTC -4h)

[ALL SESSIONS](#)

[SPEAKERS](#)

Security is Easier Before PCB Assembly: Easy Threat Modeling for Hardware

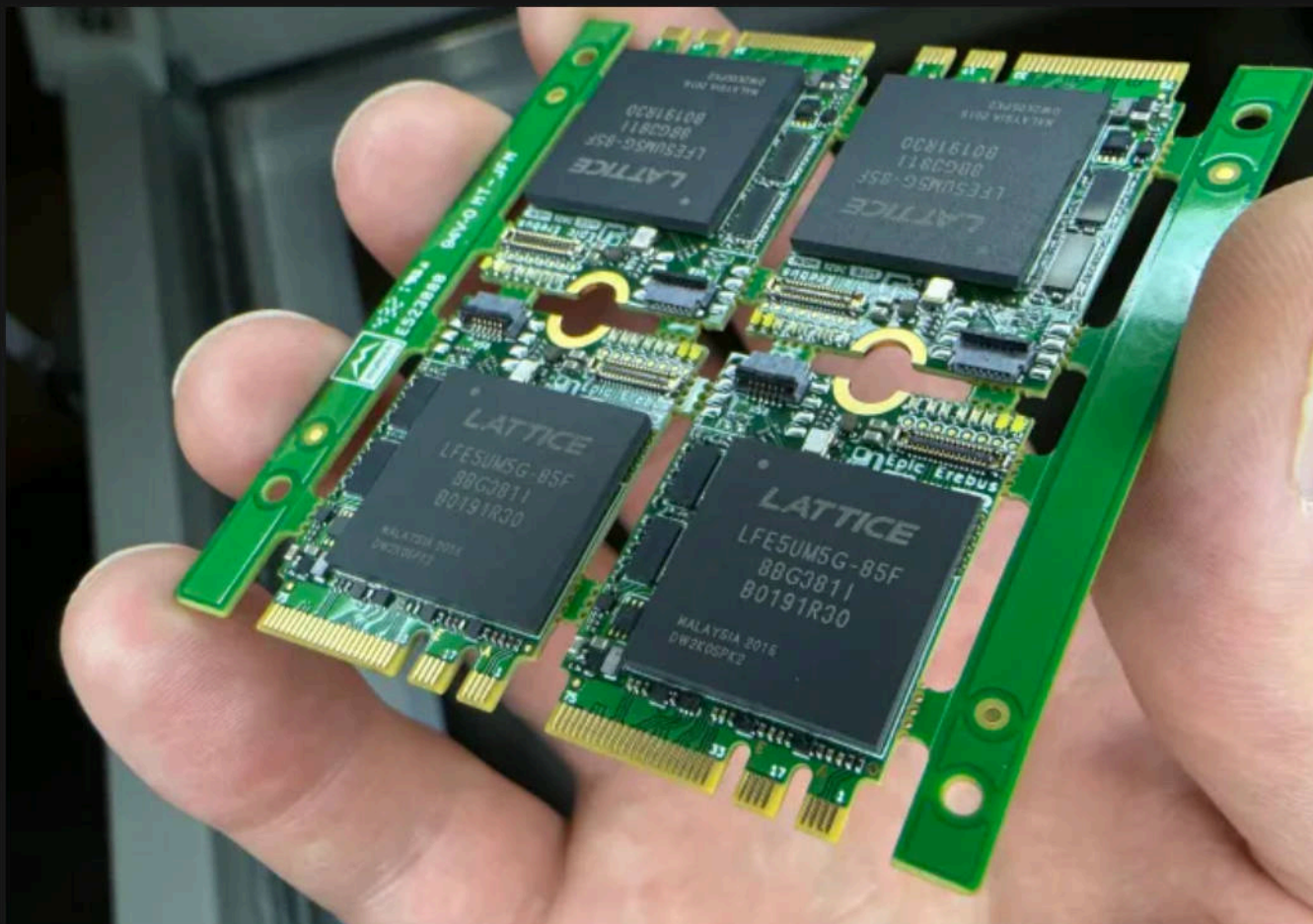
[Eric Evenchick](#) | Co-Founder and Managing Partner, Tetrel Security

[Adam Shostack](#) | President, Shostack + Associates

[Joe FitzPatrick](#) | Trainer and Researcher, SecuringHardware.com

Format: 45-Minute Briefings

Tracks: Security Essentials & Lessons Learned, Cyber-physical & Embedded Security



THANKS

Prior Art/Inspiration:

- Kingpin
- Metlstorm
- Snare
- Carmaa
- Ufrisk
- Mossman

Support/tooling

- Esden
- Aki-Nyan
- Dragonmux
- Whitequark
- Gaya Tech
- Dean Pierce

And many more...

DIRECT MEMORY ACCESS EVERYWHERE

Presentation:

[github.com/securelyfitz/
DirectMemoryAccessEverywhere](https://github.com/securelyfitz/DirectMemoryAccessEverywhere)

Erebus Design:

github.com/epic-erebus

CrowdSupply:

crowdsupply.com/securinghw/erebus



DIRECT MEMORY ACCESS EVERYWHERE

securelyfitz & baelfire 9 AUG 2025

